

SHARATH S RAO

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SUMMARY

GenAI Systems Engineer with 2+ years building production-grade LLM systems serving 24,000+ users. Specializes in multi-agent architecture (LangGraph), RAG pipelines, context engineering, and Evaluation-Driven Development. Proven track record shipping reliable AI systems using CI-integrated eval pipelines, LLM-as-Judge, and structured quality gates. Equally comfortable across the full stack — AI systems, backend engineering, and QA — in early-stage product environments. Open-source contributor to the LangChain ecosystem.

TECHNICAL SKILLS

LLM Orchestration: LangChain, LangGraph, LangSmith, Ollama
Multi-Agent Systems: Multi-Agent Architecture, Agentic Workflows, Graph-based Workflow Orchestration
Evaluation & Reliability: Evaluation-Driven Development (EDD), LLM-as-Judge, RAGAS, Promptfoo, DeepEval, CI-integrated Evals, Hallucination Detection, Guardrails, Red Teaming, Output Validation
RAG & Retrieval: Retrieval-Augmented Generation, Semantic Search, Hybrid Search, MMR, HNSW, Embeddings, Chunking Strategies
Vector Databases: PostgreSQL pgvector, Pinecone
Prompt & Context Engineering: Prompt Engineering, Context Engineering, Structured Outputs, Function Calling, Tool Calling, Schema-Constrained Prompting
Open-Source Models: DeepSeek, Llama, local model deployment via Ollama
Performance: Batching, Parallel Processing, Async Pipelines, Model Routing, Retry/Fallback, Low-Latency Inference
Full Stack & Infra: Python, NestJS, Next.js, REST APIs, PostgreSQL, Docker, System Design (HLD/LLD), UML, Multi-tenancy

EXPERIENCE

- | | |
|--|---|
| Technical Head — AI, Full Stack & QA
<i>TestMySkills.ai · Zyni Innovations pvt ltd</i> | Jan 2025 – Present
Bangalore, India |
| <ul style="list-style-type: none">Led architecture, engineering, and delivery of a B2C/B2B skill assessment platform across AI, full-stack, and QA — scaling to 24,000+ users with an 8-engineer team.Pioneered Evaluation-Driven Development (EDD) as a CI-integrated practice, enabling confident feature shipping 95%+ of the time. | |
| AI Engineer
<i>Zysk Technologies</i> | Jun 2024 – Jan 2025
Bangalore, India |
| <ul style="list-style-type: none">Built the AI-based Skill Mastery System serving 24,000+ users; reduced evaluation latency 80% through batching and parallel LLM execution.Engineered fault-tolerant LLM inference with routing, retry logic, fallback chains, and LLM-as-Judge quality pipelines. | |
| AI Engineer Intern
<i>Zysk Technologies</i> | Mar 2023 – Jun 2024
Bangalore, India |
| <ul style="list-style-type: none">Built prompt engineering frameworks, structured-output pipelines, and hallucination mitigation strategies across multiple product use cases. | |

PROJECTS

- | | |
|---|--------------------|
| TestMySkills.ai — B2C/B2B Skill Assessment Platform
<i>Zyni Innovations · Technical Head — AI, Full Stack & QA</i> | Jan 2025 – Present |
| <ul style="list-style-type: none">AI Systems: Architected multi-agent LLM systems using LangGraph with modular, graph-based workflow orchestration for intelligent, adaptive assessment delivery.Evaluation-Driven Development: Built a CI-integrated eval pipeline where known failure cases are stored in a DB and automatically replayed as pre-commit checks — enabling confident feature shipping 95%+ of the time.RAG Pipeline: Achieved ~90% accuracy using advanced chunking, embedding tuning, and MMR retrieval; monitored with LangSmith observability. | |

- **Semantic Search:** Optimized with PostgreSQL pgvector and HNSW indexing, improving retrieval relevance and downstream LLM response quality.
- **Full Stack:** Built and maintained B2C and B2B features across Next.js frontend and NestJS backend with multi-tenant architecture for enterprise clients.
- **QA Engineering:** Established quality processes, test coverage strategies, and automated testing pipelines across the platform.

Zyskie — AI-Powered HRMS Platform

2026

Zyni Innovations · Product Owner & Lead Engineer · Team of 8

- Built a full GenAI-powered HRMS covering intelligent hiring, workforce management, and internal talent discovery.
- Owned **end-to-end system architecture** (HLD + LLD + UML diagrams) and led technical delivery across the full stack.
- Implemented **PostgreSQL pgvector with HNSW and MMR** for embedding-based semantic search — enabling expert identification, mentor matching, and AI-driven team formation.
- Deployed **open-source models (DeepSeek) via Ollama** for cost-effective, privacy-preserving local inference.
- Built LLM-powered hiring workflows: JD generation, candidate evaluation, and automated scoring pipelines.
- Integrated RBAC, n8n workflow automation, and SaaS connectors (Zoho, Keka, BambooHR) on a PostgreSQL backend.

PushsendAI — Intelligent Email Marketing Platform

2024 – 2025

Zysk Technologies · AI Engineer

- Designed and built a **personalized product recommendation engine** using reinforcement learning and matrix factorization on user purchase histories and profiles.
- Engineered **prompt-based email personalization** using open-source models for dynamic content generation at scale.
- Recommendation system validated and commended by **Veronica Beard** (US fashion brand) during pilot evaluation.

EDUCATION

B.Tech in Information Science & Engineering

2020 – 2024

JSS Academy of Technical Education

Bangalore, India

- CGPA: 8.8 / 10.0

PUBLICATIONS

Hardware Inventory Management System Using IoT — IEEE (9 citations)

Interactive Framework for Querying Data from Large PDFs — IEEE (7 citations)

Traffic Control Management System for Emergency Vehicles — Telematique Journal

OPEN SOURCE

LangChain (langchain-ai) — Active contributor to the LangChain open-source ecosystem; contributed to the HANAVector self-query retriever module through hands-on internals review.

SPEAKING & COMMUNITY

Keynote Speaker — IEEE Student Chapter Inauguration, Bangalore (150+ attendees); spoke on the intersection of GenAI and metacognition in learning and knowledge systems.

Prompt Engineering Educator — Conducted 5+ sessions at KSIT, JSS, SJCE Mysore, Presidency University for student and faculty development programs; 30–50 attendees per session.

Head of Advisory Committee — Basavanagudi Government School.

AWARDS & RECOGNITION

Emerging Leader Award (2025) — Recognized for technical leadership and platform impact.

Captain of Change Agents — Zysk Technologies.